

PALLAVI KAILAS

+91 8086484000

kailaspallavi464@gmail.com

linkedin/pallavikailas

github/pallavikailas

SUMMARY

Engineer building production AI systems that survive contact with real users. Experience with GenAI backends at **Google**, and demand forecasting at **EY**. ICLR-accepted research, Smart India Hackathon qualifier, led IAESTE India to rank #2 globally during my term as National Secretary. Strongest in LLM & RAG engineering, applied ML, deployment, and backend services.

EDUCATION

Manipal Institute of Technology

B.Tech, Computer Science & Engineering (AI & ML) | Minor: Financial Technology

Aug 2022 – May 2026

CGPA: 7.7/10

Relevant Coursework: Distributed Systems, Parallel Programming, Reinforcement Learning, Database Systems, Machine Learning, Deep Learning, Big Data Analytics

WORK EXPERIENCE

EY India

Data Scientist Intern, AI & Data

PySpark • Azure Databricks • XGBoost • LSTM • Time-Series Forecasting • MLOps

Feb 2026 – Present

Bengaluru, IN

- Cut incident resolution time 50% by debugging a P1 ticket regarding distributed PySpark pipeline on Azure Databricks; root-caused systematic underforecasting to missing upstream promotional data
- Shipped 104-week demand forecasts across 5 FMCG categories against weekly SLAs on production PySpark pipelines

Virallens

AI Research Engineer Intern

LangGraph • LangChain • Milvus • FastAPI • Docker • Vector Databases

Jan 2026

Bengaluru, IN

- Cut legal-document review time 60% by engineering a metadata-extraction RAG pipeline with LangGraph + Milvus
- Improved long-document retrieval precision 25% on internal benchmarks via a semantic retriever with clustering and re-ranking
- Shipped a containerised FastAPI retrieval microservice to production, hardened against data-poisoning and injection attacks

Google

Software Engineer Intern

Go • Python • Gemini 1.5 Flash • GCP • gRPC • Retrieval-Augmented Generation

May – Jul 2025

Bengaluru, IN

- Lifted GCP Database Center Chat query relevance by 40% by integrating retrieval-augmented contextual prompts with Gemini 1.5 Flash
- Shipped Go and Python services across 3 GCP services with 100% code coverage, achieving end-to-end production data retrieval

IBS Software

Project Trainee

Autoencoders • SHAP • LIME • Anomaly Detection • Explainable AI

May – Jun 2024

Thiruvananthapuram, IN

- Reduced false-positive anomaly flags 20% across airline operations systems serving 12,000+ users using an autoencoder with SHAP/LIME interpretability across 3 LLM-backed systems

PROJECTS

FairLens | FastAPI, GCP Cloud Run, Terraform, BigQuery Google Solution Challenge 2026

Apr 2026

- Deployed a model-agnostic bias-audit platform with zero-downtime rollouts via Terraform-provisioned GCP, multi-stage Docker, and Workload Identity Federation; streamed live agent activity via SSE

Sorting Algorithm Visualiser | JavaScript, Kubernetes, Jest, GitHub Actions

Mar – May 2026

- Containerised and deployed an 8-algorithm visualiser to Kubernetes with rolling updates; CI gates with 64+ Jest tests and Lighthouse audits (Perf ≥ 85 , A11y ≥ 90)

Bilingformer | PyTorch, mT5, NLP

Nov 2023 – Feb 2024

- Fine-tuned mT5 on Hinglish dialogue for 8-class sentiment classification; handled imbalance with weighted sampling and ensembling

Legal AI Assistant | BERT, NLP Smart India Hackathon Qualifier

Oct 2023

- BERT summarisation + NLU chatbot cutting legal-document review from 30 min to under 5 min

RESEARCH & LEADERSHIP

Mars Rover Manipal

Feb 2023 – Jul 2025

AI Researcher

Manipal, IN

- Co-authored “Solar Panel Segmentation: Self-Supervised Learning for Imperfect Datasets” - accepted at ICLR (A*, Tiny Paper track); method achieved IoU 0.8911 on noisy imagery

IAESTE India

Jul 2024 – Jul 2025

National Secretary

- Led national wing of a global NPO spanning 85+ countries, achieving global ranking #2 in technical exchanges; represented India as Mandated Delegate at the 78th Annual Conference

TECHNICAL SKILLS

Languages: Go Python C++ C Java SQL JavaScript Bash

ML & Data: PyTorch LangChain LangGraph FastAPI Hugging Face PySpark Milvus NumPy Pandas

Infrastructure: Docker Kubernetes Terraform GitHub Actions gRPC Linux

Cloud: GCP Cloud Run BigQuery Cloud Storage Workload Identity Azure Databricks ADLS Data Factory

Domains: RAG Distributed Systems Backend Engineering MLOps NLP